

Docker - Handling containers and images

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ sudo service docker start
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
e6590344b1a5: Pull complete
Digest: sha256:940c619fbd418f9b2b1b63e25d8861f9cc1b46e3fc8b018ccfe8b78f19b8cc4f
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker image ls
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
dpage/pgadmin4      latest          46af92358e2e    2 months ago   524MB
postgres            16-alpine      2875f9e036c2    3 months ago   275MB
hello-world         latest         74cc54e27dc4    4 months ago   10.1kB
jupyter/pyspark-notebook  ubuntu-18.04   96dae139b62b    4 years ago    4.21GB
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run ubuntu:latest
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
d9d352c11bbd: Pull complete
Digest: sha256:b59d21599a2b151e23eea5f6602f4af4d7d31c4e236d22bf0b62b86d2e386b8f
Status: Downloaded newer image for ubuntu:latest
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

Run in 1st terminal:

```
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it ubuntu:latest
root@c9bfb29843b:/#
```

Then run in 2nd terminal:

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
c9bfb29843b   ubuntu:latest  "/bin/bash"             8 seconds ago Up 7 seconds
d6cc0b428d28   postgres:16-alpine  "docker-entrypoint.s..." 6 weeks ago   Up 11 minutes  0.0.0.0:5432->5432/tcp, :::5432->5432/tcp
475951b12750   pg_container    "docker-entrypoint.s..." 6 weeks ago   Up 11 minutes  443/tcp, 0.0.0.0:5050->80/tcp, :::5050->80/tcp
pgadmin4_container
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

Switch back to 1st terminal:

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it ubuntu:latest
root@c9bfba29843b:/# exit
exit
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

Switch back to 2nd terminal:

```
● ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/docker/
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
c9bfba29843b   ubuntu:latest                       "/bin/bash"             8 seconds ago Up 7 seconds
d6cc0b428d28   postgres:16-alpine                 "docker-entrypoint.s..." 6 weeks ago   Up 11 minutes 0.0.0.0:5432->5432/tcp, :::5432->54
32/tcp
475951b12750   dpape/pgadmin4                      "/entrypoint.sh"        6 weeks ago   Up 11 minutes 443/tcp, 0.0.0.0:5050->80/tcp, :::5
050->80/tcp
pgadmin4_container
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
d6cc0b428d28   postgres:16-alpine                 "docker-entrypoint.s..." 6 weeks ago   Up 14 minutes 0.0.0.0:5432->5432/tcp, :::5432->5432
/tcp
pg_container
475951b12750   dpape/pgadmin4                      "/entrypoint.sh"        6 weeks ago   Up 14 minutes 443/tcp, 0.0.0.0:5050->80/tcp, :::505
0->80/tcp
pgadmin4_container
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

List all containers:

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls --all
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
c9036bb31b52   hello-world                         "/hello"                7 minutes ago Exited (0) 7 minutes ago
dreamy_williamson
c9bfba29843b   ubuntu:latest                       "/bin/bash"             14 minutes ago Exited (0) 11 minutes ago
cool_diffie
d5de09f43b36   ubuntu:latest                       "/bin/bash"             17 minutes ago Exited (0) 17 minutes ago
elastic_edison
d9ac90e96658   ubuntu:latest                       "/bin/bash"             18 minutes ago Exited (0) 18 minutes ago
dazzling_bell
cb5e0f4d4aa1   hello-world                         "/hello"                22 minutes ago Exited (0) 22 minutes ago
confident_bell
d6cc0b428d28   postgres:16-alpine                 "docker-entrypoint.s..." 6 weeks ago   Up 25 minutes 0.0.0.0:5432->5432/tcp
pg_container
, :::5432->5432/tcp
475951b12750   dpape/pgadmin4                      "/entrypoint.sh"        6 weeks ago   Up 25 minutes 443/tcp, 0.0.0.0:5050->
80/tcp, :::5050->80/tcp
pgadmin4_container
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

Start container and stop container:

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container start -a dreamy_williamson
Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container stop dreamy_williamson
dreamy_williamson
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

Run in 1st terminal:

```
● ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/docker/
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it ubuntu:latest bash
root@f07a703de8de:/#
```

Run in 2nd terminal:

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
f07a703de8de   ubuntu:latest   "bash"                  17 seconds ago    Up 16 seconds
d6cc0b428d28   nice_chatterjee "docker-entrypoint.s..." 6 weeks ago      Up 35 minutes    0.0.0.0:5432->5432/tcp, :::5432->5432/tcp
432/tcp
475951b12750   postgres:16-alpine "docker-entrypoint.s..." 6 weeks ago      Up 35 minutes    443/tcp, 0.0.0.0:5050->80/tcp, :::5050->80/tcp
pg_container
pgadmin4
pgadmin4_container
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
f07a703de8de   ubuntu:latest   "bash"                  4 minutes ago    Up 4 minutes
d6cc0b428d28   nice_chatterjee "docker-entrypoint.s..." 6 weeks ago      Up 39 minutes    0.0.0.0:5432->5432/tcp, :::5432->5432/tcp
32/tcp
475951b12750   postgres:16-alpine "docker-entrypoint.s..." 6 weeks ago      Up 39 minutes    443/tcp, 0.0.0.0:5050->80/tcp, :::5050->80/tcp
pg_container
pgadmin4
pgadmin4_container
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container stop dreamy_williamson
dreamy_williamson
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
f07a703de8de   ubuntu:latest   "bash"                  5 minutes ago    Up 5 minutes
c9036bb31b52   hello-world   "/hello"                22 minutes ago    Exited (0) 2 minutes ago
c9bfba29843b   dreamy_williamson "bin/bash"              29 minutes ago    Exited (0) 26 minutes ago
d5de09f43b36   cool_diffie   "/bin/bash"             32 minutes ago    Exited (0) 32 minutes ago
d9ac90e96658   elastic_edison "/bin/bash"             33 minutes ago    Exited (0) 33 minutes ago
cb5e0f4d4aa1   dazdling_bell "/bin/bash"             37 minutes ago    Exited (0) 37 minutes ago
d6cc0b428d28   confident_bell "docker-entrypoint.s..." 6 weeks ago      Up 40 minutes    0.0.0.0:5432->5432/tcp, :::5432->5432/tcp
pg_container
475951b12750   postgres:16-alpine "docker-entrypoint.s..." 6 weeks ago      Up 40 minutes    443/tcp, 0.0.0.0:5050->80/tcp, :::5050->80/tcp
pgadmin4
pgadmin4_container
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --detach --name my_ubuntu ubuntu:latest bash
8b89305b4ff0c6e84d858ea637fa7d2aa992397b89a299fe92d26f88450e2890
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container stop my_ubuntu
my_ubuntu
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container rm my_ubuntu
my_ubuntu
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm --detach --name my_ubuntu ubuntu:latest bash
065ee246ff7bd43741a7b858f9e0ef09436ab1a21da21c8bf7f4b225c4a6f25e
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container stop my_ubuntu
Error response from daemon: No such container: my_ubuntu
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls --all
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
f07a703de8de   ubuntu:latest  "bash"                   13 minutes ago Up 13 minutes
c9036bb31b52   hello-world    "/hello"                 29 minutes ago Exited (0) 9 minutes ago
c9bfba29843b   ubuntu:latest  "/bin/bash"              37 minutes ago Exited (0) 34 minutes ago
d5de09f43b36   ubuntu:latest  "/bin/bash"              40 minutes ago Exited (0) 40 minutes ago
d9ac90e96658   ubuntu:latest  "/bin/bash"              40 minutes ago Exited (0) 40 minutes ago
cb5e0f4d4aa1   hello-world    "/hello"                 45 minutes ago Exited (0) 45 minutes ago
d6cc0b428d28   postgres:16-alpine "docker-entrypoint.s..." 6 weeks ago   Up 48 minutes    0.0.0.0:5432->5432/tcp
, :::5432->5432/tcp
pg_container
475951b12750   dpkg:pgadmin4   "/entrypoint.sh"         6 weeks ago   Up 48 minutes    443/tcp, 0.0.0.0:5050->
>80/tcp, :::5050->80/tcp
pgadmin4_container
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -it --rm --name my_ubuntu -e "my_variable=
'hello everyone'" ubuntu:latest bash
root@c41aa5ac60b0:/# echo $my_variable
'hello everyone'
root@c41aa5ac60b0:/# exit
exit
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -d --name my_ubuntu ubuntu:latest
4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container inspect my_ubuntu
[
  {
    "Id": "4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee",
    "Created": "2025-06-13T15:28:28.359289848Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "exited",
      "Running": false,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 0,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2025-06-13T15:28:28.730835326Z",
      "FinishedAt": "2025-06-13T15:28:28.732741081Z"
    },
    "Image": "sha256:bf16bdcff9c96b76a6d417bd8f0a3abe0e55c0ed9bdb3549e906834e2592fd5f",
    "ResolvConfPath": "/var/lib/docker/containers/4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee/hostname",
    "HostsPath": "/var/lib/docker/containers/4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee/hosts",
    "LogPath": "/var/lib/docker/containers/4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee/4c20374f2989c840ded08547d6b5452bb829ee49d2c9a0ce9eb9cd474f93ecee-json.log",
    "Name": "/my_ubuntu",
    "RestartCount": 0,
    "Driver": "overlay2",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "docker-default",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
        "Config": {}
      },
      "NetworkMode": "default",
      "PortBindings": {},
      "RestartPolicy": {
        "Name": "no",
        "MaximumRetryCount": 0
      },
      "AutoRemove": false,
      "VolumeDriver": "",
      "VolumesFrom": null,
      "ConsoleSize": [
        56,
        128
      ],
      "CapAdd": null,
      "CapDrop": null,
      "CgroupsMode": "host",
      "Dns": [],
      "DnsOptions": [],
      "DnsSearch": [],
      "ExtraHosts": null,
      "GroupAdd": null,
      "IpcMode": "private",
      "Cgroup": "",
      "Links": null,
      "OomScoreAdj": 0,
      "PidMode": ""
    }
  }
]

```

```
"OomScoreAdj": 0,
"PidMode": "",
"Privileged": false,
"PublishAllPorts": false,
"ReadOnlyRootfs": false,
"SecurityOpt": null,
"UTSMode": "",
"UsersnsMode": "",
"ShmSize": 67108864,
"Runtime": "runc",
"Isolation": "",
"CpuShares": 0,
"Memory": 0,
"NanoCpus": 0,
"CgroupParent": "",
"BlkioWeight": 0,
"BlkioWeightDevice": [],
"BlkioDeviceReadBps": [],
"BlkioDeviceWriteBps": [],
"BlkioDeviceReadIops": [],
"BlkioDeviceWriteIops": [],
"CpuPeriod": 0,
"CpuQuota": 0,
"CpuRealtimePeriod": 0,
"CpuRealtimeRuntime": 0,
"CpusetCpus": "",
"CpusetMems": "",
"Devices": [],
"DeviceCgroupRules": null,
"DeviceRequests": null,
"MemoryReservation": 0,
"MemorySwap": 0,
"MemorySwappiness": null,
"OomKillDisable": false,
"PidLimit": null,
"Ulimits": [],
"CpuCount": 0,
"CpuPercent": 0,
"IOMaximumIops": 0,
"IOMaximumBandwidth": 0,
"MaskedPaths": [
  "/proc/asound",
  "/proc/acpi",
  "/proc/kcore",
  "/proc/keys",
  "/proc/latency_stats",
  "/proc/timer_list",
  "/proc/timer_stats",
  "/proc/sched_debug",
  "/proc/scsi",
  "/sys/firmware",
  "/sys/devices/virtual/powercap"
],
"ReadonlyPaths": [
  "/proc/bus",
  "/proc/fs",
  "/proc/iq",
  "/proc/sys",
  "/proc/sysrq-trigger"
]
},
"GraphDriver": {
  "Data": {
    "LowerDir": "/var/lib/docker/overlay2/060988410fb5c05ce90bde2f15b9d8946ac37c1ece9da5e5e308a2a36dbcb4f2-init/diff:/var/lib/docker/overlay2/f0ab3da513c8043e249a85d2f0b1c79bdaa28156c5d71cb80ac44d14fc734b12/diff",
    "MergedDir": "/var/lib/docker/overlay2/060988410fb5c05ce90bde2f15b9d8946ac37c1ece9da5e5e308a2a36dbcb4f2/merged",
    "UpperDir": "/var/lib/docker/overlay2/060988410fb5c05ce90bde2f15b9d8946ac37c1ece9da5e5e308a2a36dbcb4f2/diff",
    "WorkDir": "/var/lib/docker/overlay2/060988410fb5c05ce90bde2f15b9d8946ac37c1ece9da5e5e308a2a36dbcb4f2/work"
  }
}
```

```

    "WorkDir": "/var/lib/docker/overlay2/060988410fb5c05ce90bde2f15b9d8946ac37c1ece9da5e5e308a2a36dbcb4f2/work"
  },
  "Name": "overlay2"
},
"Mounts": [],
"Config": {
  "Hostname": "4c20374f2989",
  "Domainname": "",
  "User": "",
  "AttachStdin": false,
  "AttachStdout": false,
  "AttachStderr": false,
  "Tty": false,
  "OpenStdin": false,
  "StdinOnce": false,
  "Env": [
    "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
  ],
  "Cmd": [
    "/bin/bash"
  ],
  "Image": "ubuntu:latest",
  "Volumes": null,
  "WorkingDir": "",
  "Entrypoint": null,
  "OnBuild": null,
  "Labels": {
    "org.opencontainers.image.ref.name": "ubuntu",
    "org.opencontainers.image.version": "24.04"
  }
},
"NetworkSettings": {
  "Bridge": "",
  "SandboxID": "d3157a5465ab956bd9b5abae24f09fa6f89b5faebfbf3cfe67c7f5de2da4c519",
  "SandboxKey": "/var/run/docker/netns/d3157a5465ab",
  "Ports": {},
  "HairpinMode": false,
  "LinkLocalIPv6Address": "",
  "LinkLocalIPv6PrefixLen": 0,
  "SecondaryIPAddresses": null,
  "SecondaryIPv6Addresses": null,
  "EndpointID": "",
  "Gateway": "",
  "GlobalIPv6Address": "",
  "GlobalIPv6PrefixLen": 0,
  "IPAddress": "",
  "IPPrefixLen": 0,
  "IPv6Gateway": "",
  "MacAddress": "",
  "Networks": {
    "bridge": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": null,
      "MacAddress": "",
      "NetworkID": "81af6e80096ca4822b6cc57901cea1ce955046940c7283b4d69a9349fa649a80",
      "EndpointID": "",
      "Gateway": "",
      "IPAddress": "",
      "IPPrefixLen": 0,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "DriverOpts": null,
      "DNSNames": null
    }
  }
}
}
}

```

```

    "LinkLocalIPv6Address": "",
    "LinkLocalIPv6PrefixLen": 0,
    "SecondaryIPAddresses": null,
    "SecondaryIPv6Addresses": null,
    "EndpointID": "",
    "Gateway": "",
    "GlobalIPv6Address": "",
    "GlobalIPv6PrefixLen": 0,
    "IPAddress": "",
    "IPPrefixLen": 0,
    "IPv6Gateway": "",
    "MacAddress": "",
    "Networks": {
      "bridge": {
        "IPAMConfig": null,
        "Links": null,
        "Aliases": null,
        "MacAddress": "",
        "NetworkID": "81af6e80096ca4822b6cc57901cea1ce955046940c7283b4d69a9349fa649a80",
        "EndpointID": "",
        "Gateway": "",
        "IPAddress": "",
        "IPPrefixLen": 0,
        "IPv6Gateway": "",
        "GlobalIPv6Address": "",
        "GlobalIPv6PrefixLen": 0,
        "DriverOpts": null,
        "DNSNames": null
      }
    }
  }
}
}
}

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker\$